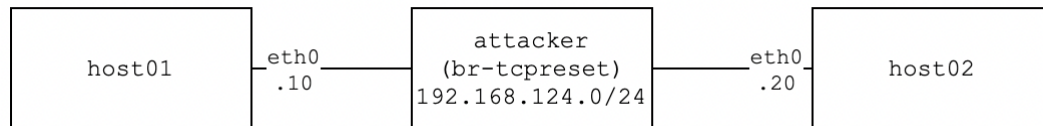


TCP Reset [40 points]

Consider the topology in the image below. The attacker has already established a *Man-in-the-middle* (MITM) presence between host 1 and host 2. This allows the attacker to see and manipulate the traffic between the two hosts. You should exploit this position to conduct a TCP-reset attack.



First, unzip the provided file and navigate to the unzipped directory. Afterward, execute the commands below to generate the topology.

```
docker compose build
docker compose up -d
```

To connect to the attacker container, execute the command below.

```
docker exec -ti assignment_4_tcp_reset_attacker bash
```

The two hosts exchange data with each other via a TCP connection. Host 2 connects to host 1 on port 1337. You can view the messages the two hosts exchange in real time via the following command:

```
docker logs assignment_4_tcp_reset_host2 --follow
```

After finishing the assignment, you can destroy the environment using the command below. Make sure your script is inside the mounted *solution* folder so it is saved when you destroy the environment.

```
docker compose down
```

Goal

Create a Python3 script which accepts three arguments. These arguments should be the source IP address, destination IP address, and destination port of the TCP connection. The way the script should be invoked is shown below.

```
python3 tcp_reset.py 192.168.124.20 192.168.124.10 1337
```

The assignment is successful if the TCP connection breaks when executing the TCP-reset attack.

Submission Instructions

Your script should run on the attacker container. You may not use external libraries or tools different from those installed on the provided *attacker* container.